Cheng (Kevin) Shao

236-333-4484 | Kevinye0907@gmail.com | Portfolio

Education

University of Waterloo — Waterloo, ON
B.A.Sc. Electrical Engineering, Honours, Co-op — *September 2025 - Expected May 2030*

Experience

Electrical Team Member — Jan. 2026 – Present
University of Waterloo Baja SAE Team — *Waterloo, ON*

- Integrating and testing a Data Acquisition (DAQ) system using Arduino, MPU6050 accelerometers, and RPM sensors
- Designing custom PCBs and developing an upgraded 18V battery setup to regulate internal electrical components
- Researching and coding strain gauges to extract live telemetry data, aiding in comprehensive system calibration
- Fabricating space-efficient wire harnesses and designing circuit schematics for 2026 competition vehicles

Project Team Member — Winter 2026
Waterloo Experience (WE) Accelerate Program — *Remote*

- Conducted accessibility and usability audits of five co-op community pages to identify UI/UX failures
- Developed digital accessibility micro-course modules focusing on accessible HTML, CSS, and web design
- Evaluated digital environments for strict AODA compliance and WCAG standard adherence

Manual Labour Worker — Dec. 2024 – Jul. 2025
Vanderkooi Farm — *Nanaimo, BC*

- Collaborated with a fast-paced team to execute large-scale livestock transfer operations
- Processed over 20,000 chickens per shift while strictly adhering to humane handling requirements
- Maintained high performance and reliability during intense, high-volume seasonal peak periods

Room Maintenance Worker — Summers 2022 – 2025
Mapleview B&B — *Nanaimo, BC*

- Managed guest accommodations and property arrangements with a high standard of cleanliness
- Handled listing coordination and guest communication, securing consistently positive reviews

Projects

RPM Monitoring Device | *ESP32, C++, HW-477 Hall Effect Sensor* — Jan. 2026 – Present

- Engineered a custom rotational speed sensing solution using an ESP32 and an HW-477 Hall Effect sensor
- Developed firmware to process analog magnetic signals and calculate accurate, real-time RPM telemetry
- Integrated the sensor array into a broader vehicle DAQ framework to optimize off-road performance

Smartwatch Sleep Monitor | *ECE 198 Design Project, Arduino, C++* — Sep. 2025 – Dec. 2025

- Prototyped a wearable Arduino UNO R4 and oximeter sleep tracker to address hospital-induced delirium
- Programmed firmware to detect sleep cycles via heart rate variability thresholds (Start < 55, End > 65 BPM)
- Implemented IoT functionality to transmit automated daily sleep reports to caregivers via Wi-Fi
- Validated system reliability, confirming accurate heart rate logging and successful email data delivery

Trades & Mechanical Projects | *VIU Trades Program* — Sep. 2024 – Feb. 2025

- **Electrical:** Fabricated extension units from technical schematics, ensuring CSA grounding compliance
- **Mechanical:** Performed automotive maintenance and brake replacements, prioritizing workshop safety
- **Woodworking:** Fabricated custom products using industrial power tools and precision finishing
- **Welding:** Utilized thermal cutting and welding to fuse metals with high structural integrity

Technical Skills

Programming/Firmware: C++, HTML, Arduino (Wiring)
Hardware & Tools: ESP32, Arduino, Accelerometers (MPU6050), Multimeters, Soldering, Power Tools
Relevant Skills: Circuit Assembly, PCB Design, Wire Harnessing, Data Acquisition (DAQ), Technical Documentation
Spoken Languages: English (Fluent), Mandarin (Fluent Speaking/Listening, Intermediate Reading/Writing)